

MACHINE PROBLEM #1

Local Git Version Control

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Section: IV - CCSAD

Environment Setup

```
ieigion@YAJI MINGW64 ~
$ git config --global user.name "Ijay F. Jaculbe"
git config --global user.email "ijaculbe.a12240882@umak.edu.ph"
cd /c/Users/ieigion/OneDrive/Desktop
mkdir Jaculbe_local_project
cd Jaculbe_local_project
git init
git branch -M main
echo "Initial code" > index.html
git add index.html
git commit -m "Initial commit"
Initialized empty Git repository in C:/Users/ieigion/OneDrive/Desktop/Jaculbe_local_project/.git/
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it
[main (root-commit) 406509a] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 index.html

ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
$
```

Tasks:

1. Create a new branch:

- Create a new branch named "**feature_x**" to isolate your experimental changes.

```
ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
$ git checkout -b feature_x
Switched to a new branch 'feature_x'

ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (feature_x)
$
```

2. Stage changes:

- Make modifications to your project files.

Stage the changes using the **git add** command.

```
ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (feature_x)
$ echo "Feature X code" >> index.html
git add index.html
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it
```

3. Commit changes:

- Commit the staged changes to your local repository using the **git commit** command. Provide a clear and concise commit message.

```
ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (feature_x)
$ git commit -m "Add feature x"
[feature_x 542d12a] Add feature x
1 file changed, 1 insertion(+)
```

4. Switch back to the main branch:

Use the `git checkout` command to switch back to the main branch.

```
ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (feature_x)
$ git checkout main
Switched to branch 'main'

ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
$
```

5. Merge changes from the feature branch:

Use the `git merge` command to merge the changes from the "feature_x" branch into the main branch. Resolve any merge conflicts that may arise.

```
ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
$ echo "Main branch code" >> index.html
git add index.html
git commit -m "Add main branch updates"
git merge feature_x
echo "Resolved code from main and feature_x" > index.html
git add index.html
git commit -m "Resolve merge conflict"
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it
[main 3d8abe5] Add main branch updates
1 file changed, 1 insertion(+)
Auto-merging index.html
CONFLICT (content): Merge conflict in index.html
Automatic merge failed; fix conflicts and then commit the result.
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it
[main c966644] Resolve merge conflict

ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
```

6. Viewing the Commit History:

- Use Git commands to view the commit history and ensure that all commits are in place.

```
ieigion@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
$ git log --oneline --graph --all
*   c966644 (HEAD -> main) Resolve merge conflict
|  \
|   * 542d12a (feature_x) Add feature x
|   * | 3d8abe5 Add main branch updates
|   |/
|   * 406509a Initial commit
```

7. Tagging Versions (Optional):

- After completing the merge, use the `git tag` command to create a tag named `v1.0` to mark the first version of your project.

```
iegi@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
$ git tag v1.0

iegi@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
$

iegi@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
$

iegi@YAJI MINGW64 ~/OneDrive/Desktop/Jaculbe_local_project (main)
```